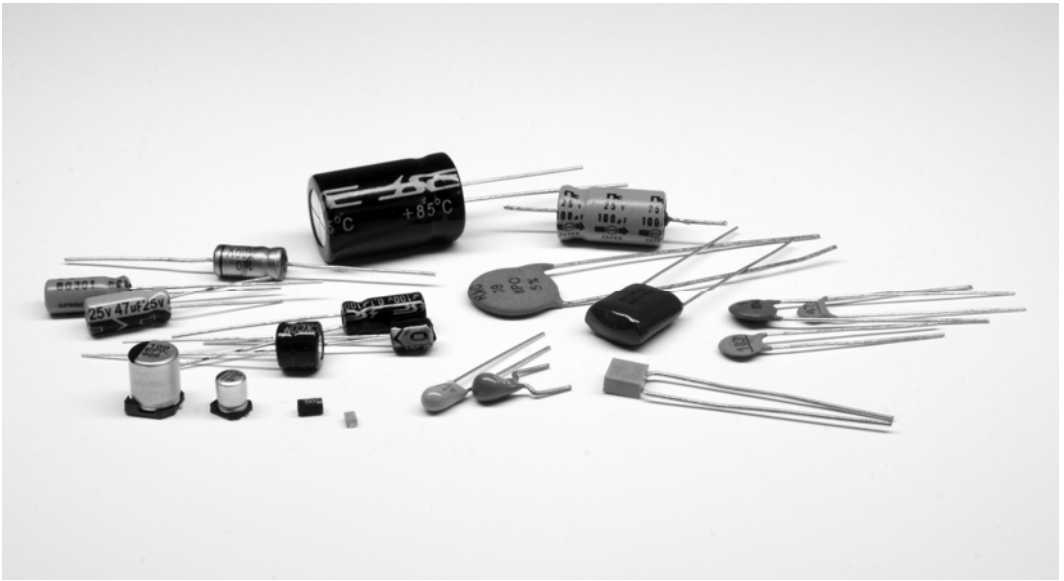


Figure 12-4. *Capacitors*

Inside a capacitor are two, parallel plates of conductive material separated by an insulating dialectic material. Common materials include an electrolytic oil, tantalum, ceramic, mica, and polyester. Some capacitors, like the electrolytic and tantalum capacitors, are polarized while others are not. Some circuits require one or the other, so pay attention to this as placing a polarized capacitor in a circuit the wrong way around might cause it to explode and damage other components.

We will often use capacitors to filter electronic signals, smoothing out any dips in the level of current caused by heavy or "noisy" loads. They are a necessary part of voltage regulation circuits and circuits that use motors to keep things running smoothly.

Diodes

Diodes are devices that will only allow the current to flow through a circuit in one direction. By necessity, diodes are polarized, meaning that they can only go into a circuit one way. The diode will only conduct electricity when the side called the anode (+) is more positive than the side called the cathode (-). The cathode in a standard diode is usually marked with a line. If current attempts to enter the diode the wrong way, it will be blocked and will not pass through. Figure 12-5 shows a selection of diodes.